Introduction to the Finite Element Method

Lecture 3: Galerkin’s Method and 1D FEM

From Last week

- **Key insight**: The weak form we saw in lecture #1 and Galerkin's method from lecture #2 are fundamentally the
- **Galerkin's approach** (review):

1. Compute the residual $Ay_N - f = r_N(x)$
2. Force the residual to be orthogonal to each approximation function:
   $\int_0^L r_N(x)\phi_i(x)dx = 0, \quad i = 1, 2, 3, \ldots N$
3. Solve the resulting set of coupled equations

However…

- The primary challenge: **no systematic way for choosing the approximation functions (yet)**
- Finite Element Method (FEM) is designed specifically to address this problem:
  - Based on Galerkin's method
  - Provides a computationally systematic and efficient approach
  - Removes restrictions on differentiability

- Creates a framework for handling complex geometries

A 1D model problem

Today we will start solving our first PDE using the finite element method (FEM).
As is often the case in real life problems, materials are not necessarily homogeneous.
Moreover, the physical solution we are looking for is not necessarily smooth.
Consider the 1D bar beshown below, composed of two different elastic solids.

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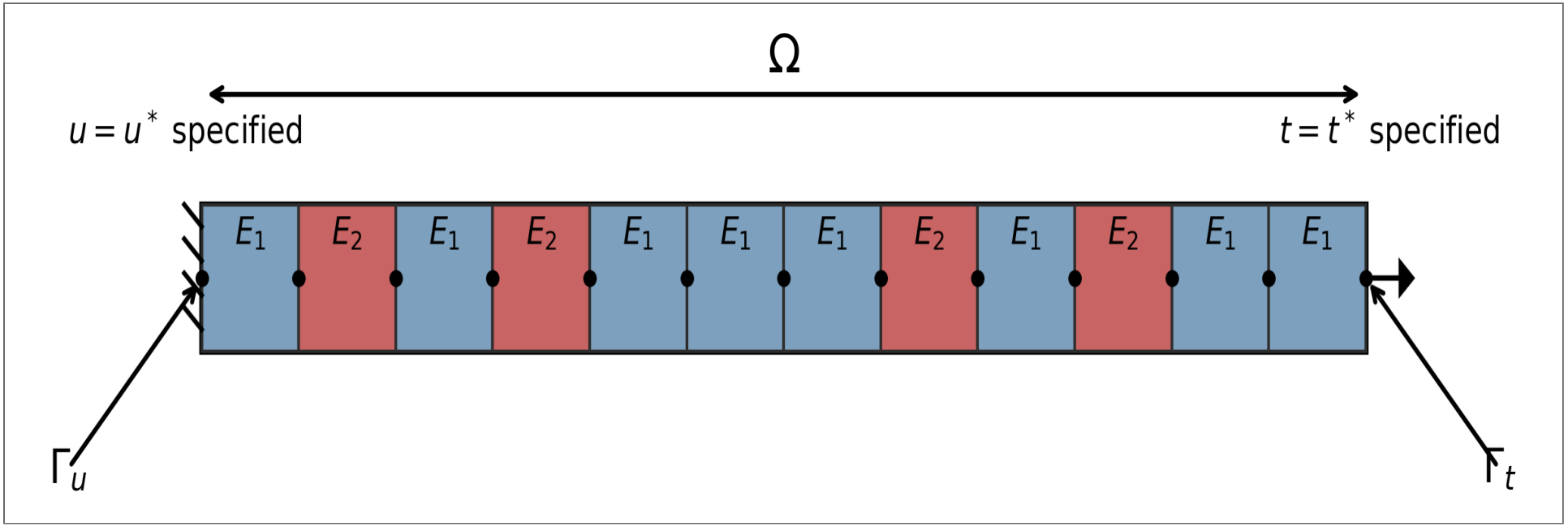

- The bar is subjected to tractions t^* on it's left side, denoted by Γ_t, while the right side is fixed, denoted by Γ_u.
- As you can see, the bar is composed of two diffeerent materials, with different Young's moduli E_1 and E_2.

A 1D model problem - Strong Form

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The strong form of the PDE:

$$f + \frac{d}{dx}\sigma(x) = 0$$

This PDE is simply a balance of forces, with f being the body force and σ the stress.

The boundary conditions:

$$u|_{\Gamma_u} = u^*$$

$$t|_{\Gamma_t} = t^*$$

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Since we assume the body to be linearly elastic, the stress σ is given by:

$$\sigma(x) = E\epsilon(x)$$

where E is the Young's modulus and ϵ is the strain.

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Recalling that for 1D and small displacements the strain is given by:

$$\epsilon(x) = \frac{d}{dx}u(x)$$

where u is the displacement.

We can now write the stress σ as:

$$\sigma(x) = E\frac{d}{dx}u(x)$$

Getting the weak form

As you recall to get the weak form we multiply by a test function v and integrate over the domain Ω.

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$$\int_0^L \left(f + \frac{d}{dx}\sigma(x) \right) v(x)\, dx = 0$$

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Using integration by parts:

$$\frac{d}{dx}\sigma(x)v(x) = \sigma(x)\frac{d}{dx}v(x) + v(x)\frac{d}{dx}\sigma(x)$$

And thus:

$$v(x)\frac{d}{dx}\sigma(x) = -\sigma(x)\frac{d}{dx}v(x) + \frac{d}{dx}\sigma(x)v(x)$$

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We can write the weak form as:

$$\int_0^L f\,v(x)\,dx - \int_0^L \sigma(x)\frac{d}{dx}v(x)\,dx + \int_0^L \frac{d}{dx}\sigma(x)v(x)\,dx = 0$$

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Requiring the $|_{\Gamma_u} = 0$ and $\sigma|_{\Gamma_t} = t^*$:
starting from the third term

$$\int_0^L \frac{d}{dx}\sigma(x)v(x)\,dx = -\sigma(0)v(0) + \sigma(L)v(L)$$

and recalling that v is 0 on the left boundary (Γ_u):

$$\int_0^L \frac{d}{dx}\sigma(x)v(x)\,dx = t^*\,v(L)$$

We can write the weak form as:

$$\int_0^L \sigma(x) \frac{d}{dx} v(x)\, dx = t^* v(L) + \int_0^L f v(x)\, dx$$

The Weak Form

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To get the final weak form in terms of the primary unknown (the displacement u), we introduce the constitutive and kinematic relations

$$\int_0^L E\frac{d}{dx}u(x)\frac{d}{dx}v(x)\,dx = t^* v(L) + \int_0^L f\,v(x)\,dx$$

The final equation is the **Weak Form** or **Variational Formulation**:

Alternatively, we can write it as:

$\mathbb{B}(u,v) = \mathbb{F}(v)$

such that:

$\mathbb{B}(u,v) = \int_0^L \frac{dv}{dx}E\frac{du}{dx}\,dx$ - representing internal virtual work

$\mathbb{F}(v) = \int_0^L f\,v\,dx + t^* v|_{\Gamma_t}$ representing external virtual work.

This form requires less smoothness on the solution u compared to the original differential equation (strong form) and is the starting point for the Finite Element Method.

Function Spaces for Weak Form

For well-defined integrals, we need:

$$\forall v, \int_0^L \frac{dv}{dx}\sigma\, dx < \infty, \int_0^L f v\, dx < \infty$$

This leads to using Sobolev spaces (recall from previous lecture):

$$u \in H^1(\Omega) \iff \int_\Omega \frac{du}{dx}\frac{du}{dx}\, dx + \int_\Omega uu\, dx < \infty$$

And our problem becomes:

Find $u \in H^1(\Omega), u|_{\Gamma_u} = u^*$, such that $\forall v \in H^1(\Omega), v|_{\Gamma_u} = 0$:

$$\int_0^L \frac{dv}{dx}E\frac{du}{dx}\, dx = \int_0^L f v\, dx + t^* v|_{\Gamma_t}$$

Approximating the Solution

We discretize the problem using:

$$u_N \approx u(x) = \sum_{j=1}^{N} a_j \phi_j(x), \quad v = \phi_i(x)$$

This gives us a linear system:

$$[K]\{a\} = \{F\}$$

Where:

- $K_{ij} = \int_0^L \frac{d\phi_i}{dx} E \frac{d\phi_j}{dx}\, dx$
- $F_i = \int_0^L f \phi_i \, dx + t^* \phi_i|_{\Gamma_t}$

Basis Functions in FEM

The key insight of FEM is a systematic way to construct basis functions:

- Functions are built piecewise over subdomains ("elements")
- Usually low-degree polynomials
- Smooth enough to be in $H^1(\Omega)$
- Form a nodal basis where $\phi_i(x_j) = \delta_{ij}$ (Kronecker delta)
- The sum over the basis functions is given by: $\sum_{j \in \text{element}} \phi_j(x) = 1$ (partition of unity)
- Each basis function is non-zero only on elements sharing node i

Basis Functions: Linear Elements

For linear elements, we define:

$$\phi_i(x) = \begin{cases} \frac{x-x_{i-1}}{h_i}, & x_{i-1} \le x \le x_i \\ 1 - \frac{x-x_i}{h_{i+1}}, & x_i \le x \le x_{i+1} \\ 0, & \text{otherwise} \end{cases}$$

where $h_i = x_i - x_{i-1}$

The derivatives are:

$$\frac{d\phi_i}{dx} = \begin{cases} \frac{1}{h_i}, & x_{i-1} \le x \le x_i \\ -\frac{1}{h_{i+1}}, & x_i \le x \le x_{i+1} \\ 0, & \text{otherwise} \end{cases}$$

Visualization of Basis Functions

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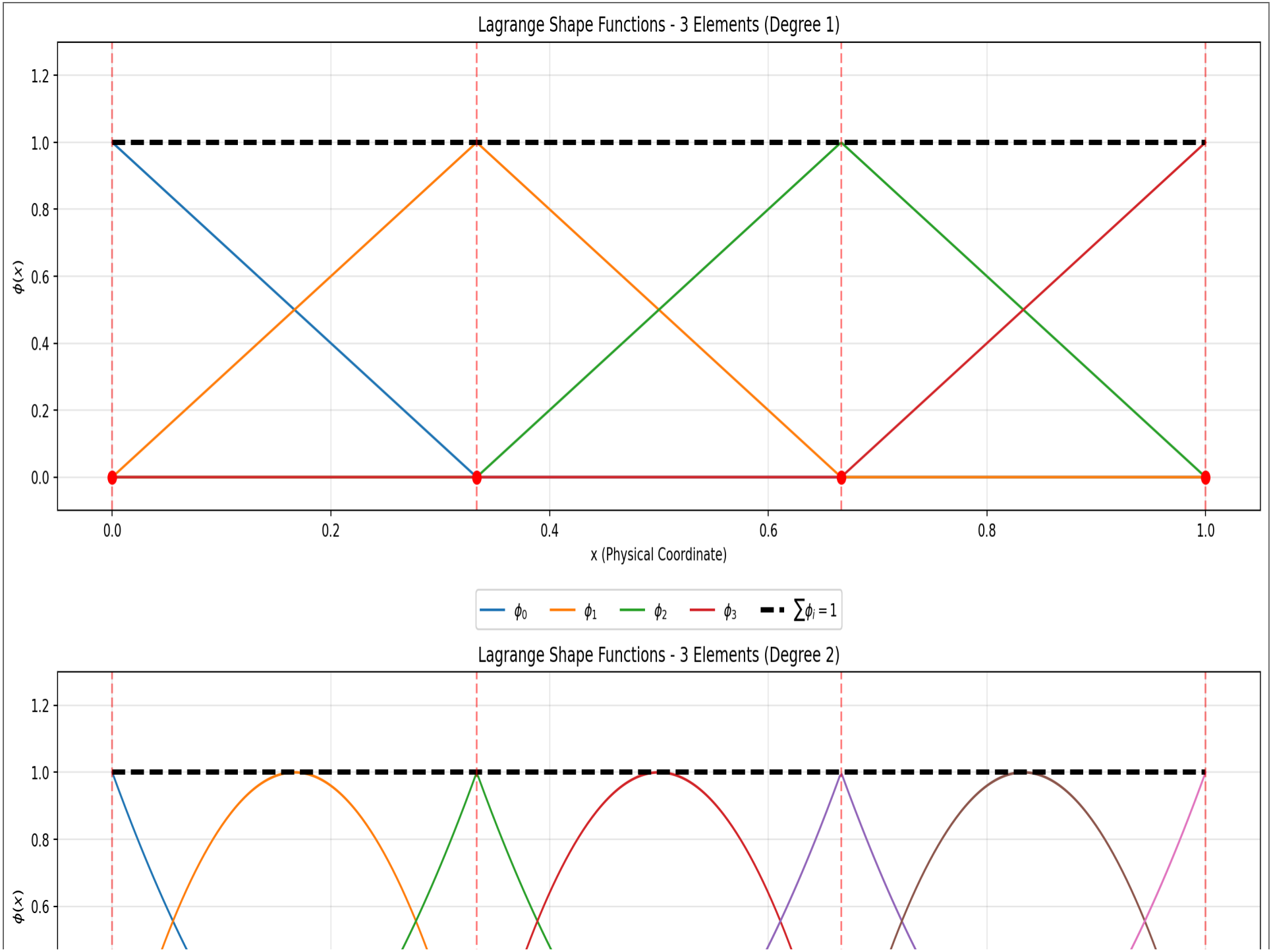
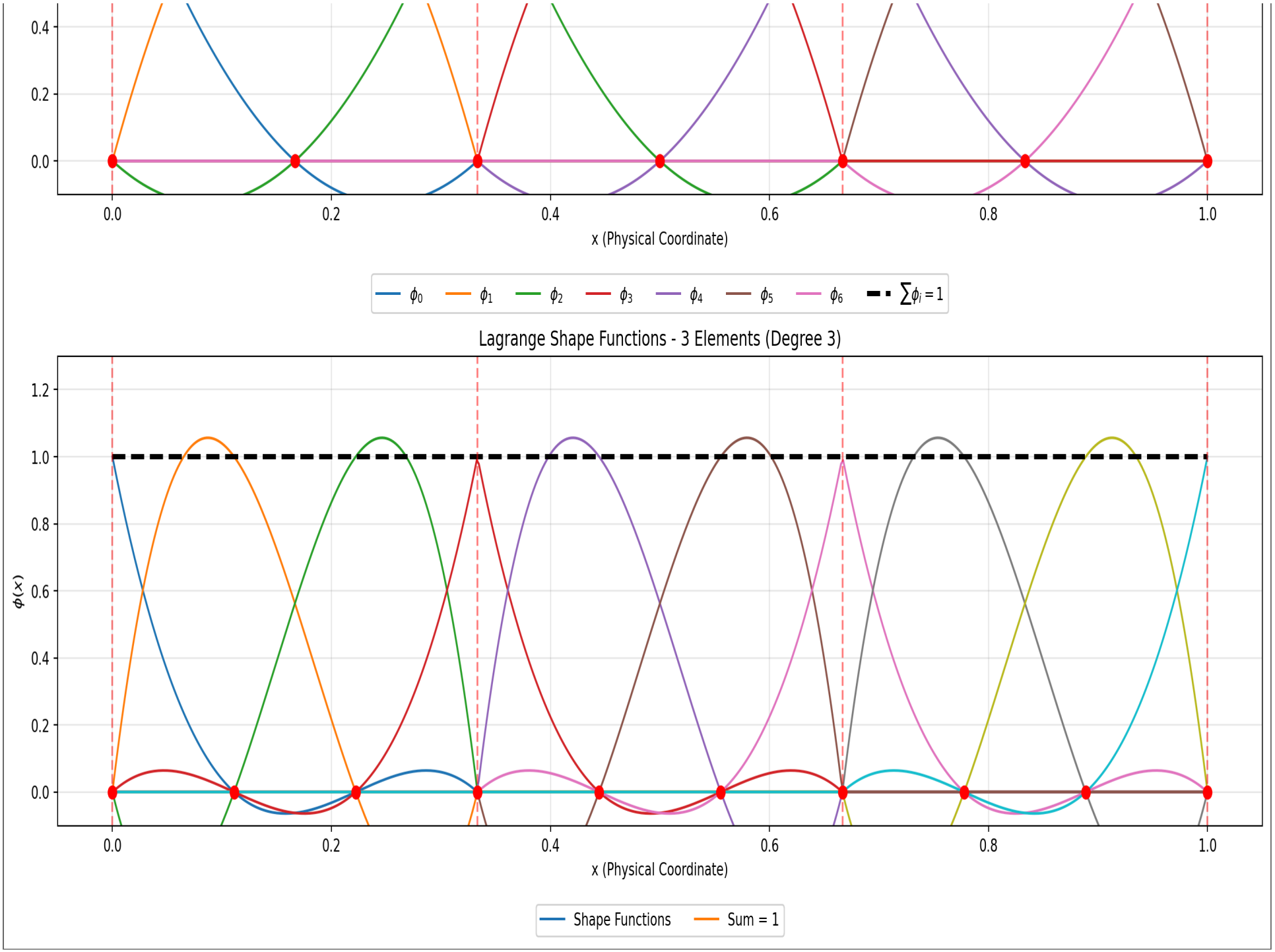
Lagrange Shape Functions


Lagrange Shape Functions - 3 Elements (Degree 1)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
$\phi(x)$
0.0
0.2
0.4
0.6
0.8
1.0
x (Physical Coordinate)
ϕ_0
ϕ_1
ϕ_2
ϕ_3
$\sum \phi_i = 1$
Lagrange Shape Functions - 3 Elements (Degree 2)
1.2
1.0
0.8
0.6
$\phi(x)$


0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
x (Physical Coordinate)
ϕ_0
ϕ_1
ϕ_2
ϕ_3
ϕ_4
ϕ_5
ϕ_6
$\sum \phi_i = 1$
Lagrange Shape Functions - 3 Elements (Degree 3)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
$\phi(x)$
0.0
0.2
0.4
0.6
0.8
1.0
x (Physical Coordinate)
Shape Functions
Sum = 1

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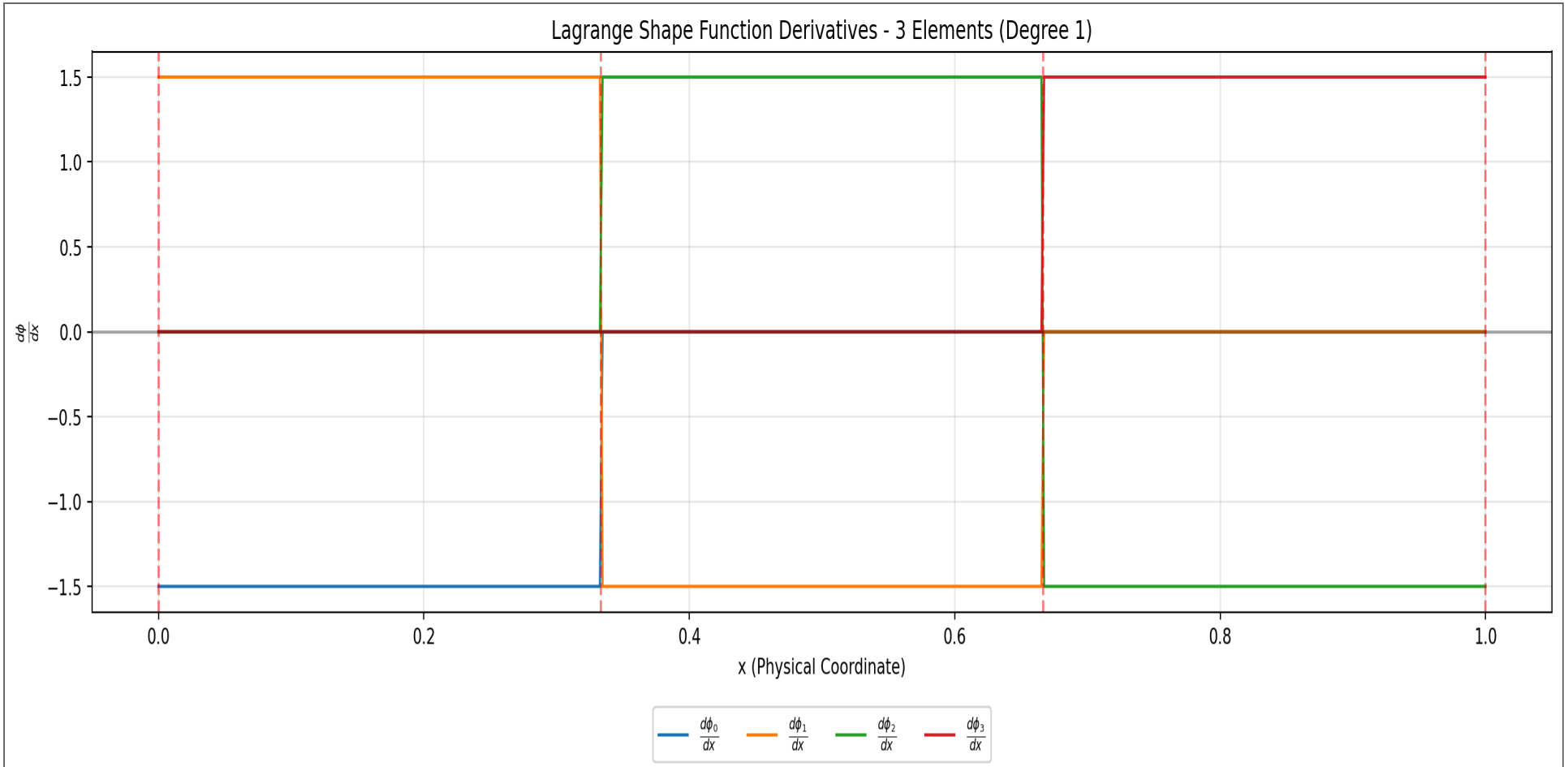
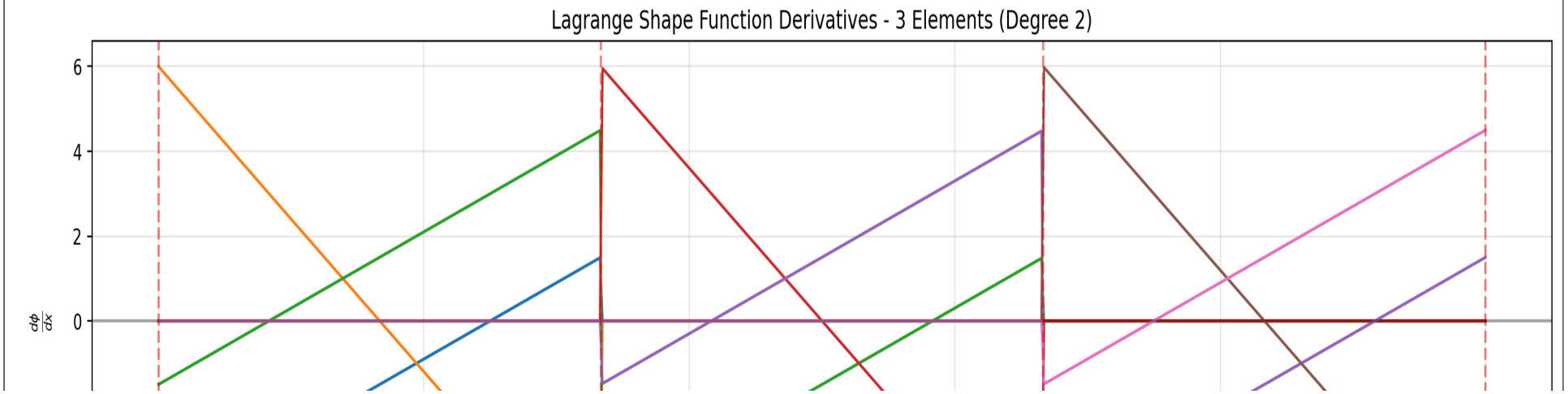
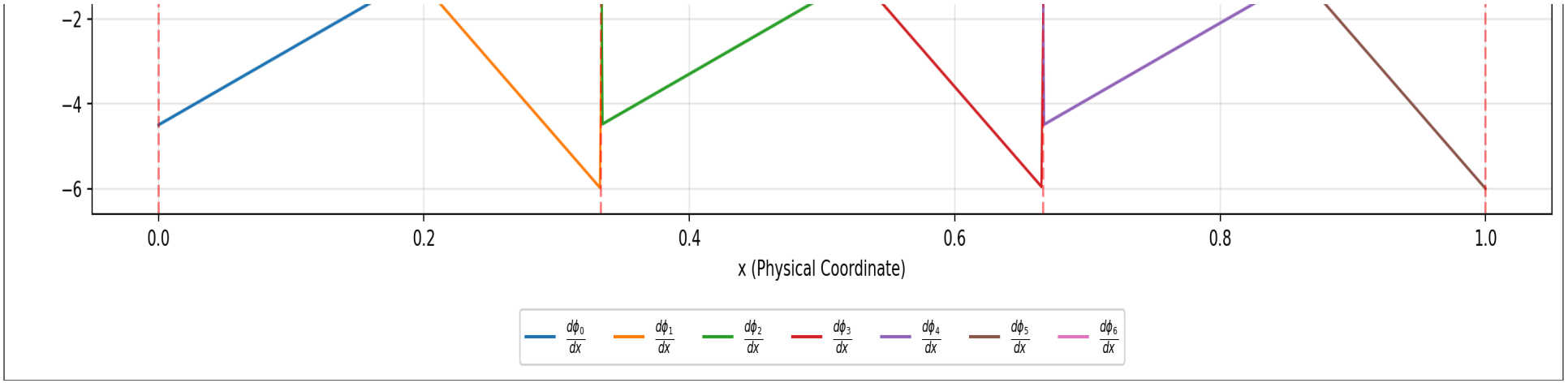
Derivatives of shape functions


Lagrange Shape Function Derivatives - 3 Elements (Degree 1)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
$\frac{d\phi}{dx}$
0.0
0.2
0.4
0.6
0.8
1.0
x (Physical Coordinate)
$\frac{d\phi_0}{dx}$
$\frac{d\phi_1}{dx}$
$\frac{d\phi_2}{dx}$
$\frac{d\phi_3}{dx}$


Lagrange Shape Function Derivatives - 3 Elements (Degree 2)
6
4
2
0
$\frac{d\phi}{dx}$



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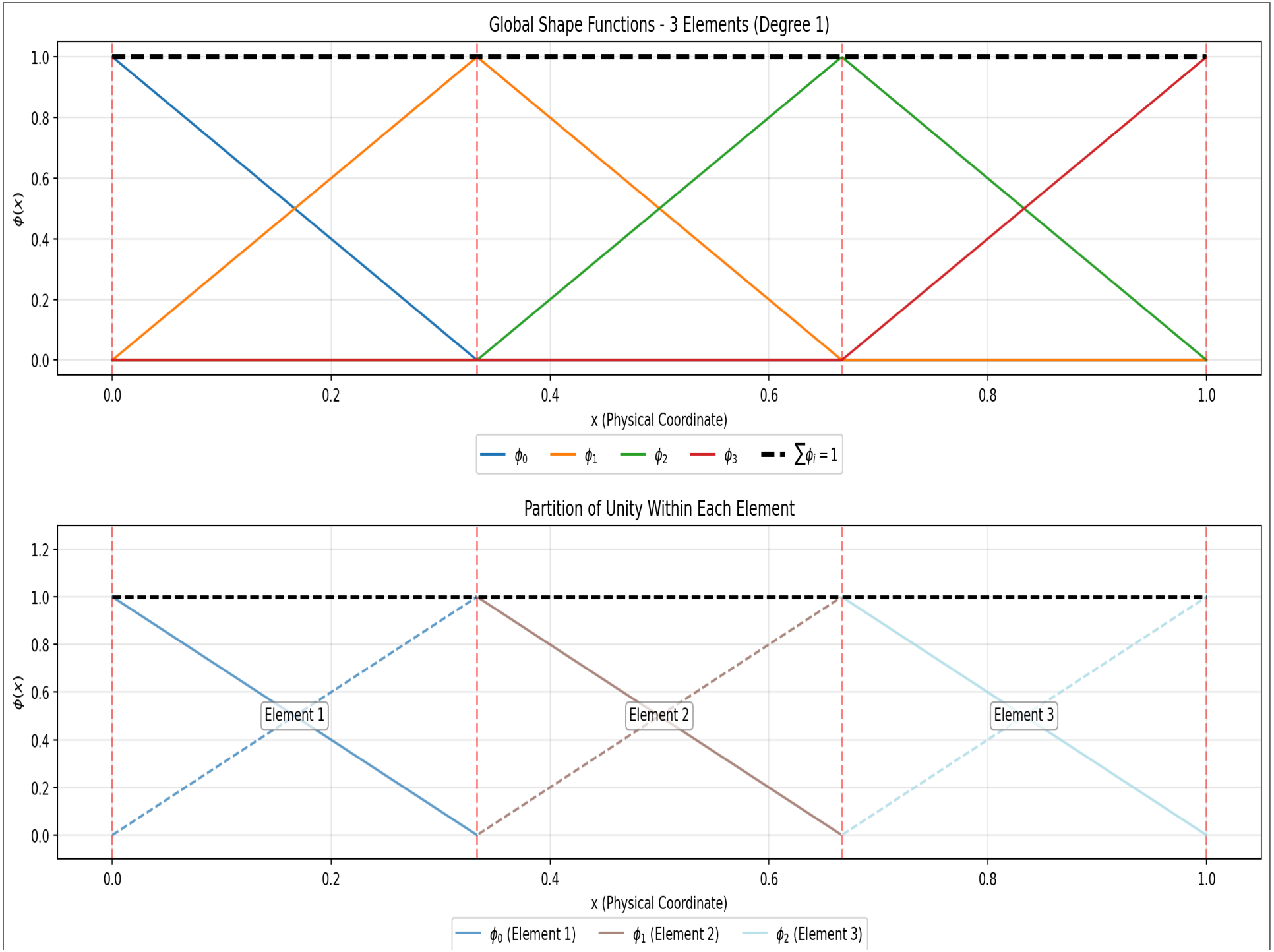
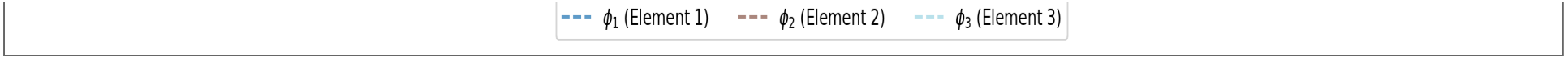
partition of unity degree 1


Global Shape Functions - 3 Elements (Degree 1)
$\phi(x)$
x (Physical Coordinate)
ϕ_0
ϕ_1
ϕ_2
ϕ_3
$\sum \phi_i = 1$
Partition of Unity Within Each Element
Element 1
Element 2
Element 3
$\phi(x)$
x (Physical Coordinate)
ϕ_0 (Element 1)
ϕ_1 (Element 2)
ϕ_2 (Element 3)


ϕ_1 (Element 1)
ϕ_2 (Element 2)
ϕ_3 (Element 3)

Building the Stiffness Matrix

For example, to compute K_{11}:

$$K_{11} = \int_0^L \frac{d\phi_1}{dx} E \frac{d\phi_1}{dx}\, dx = \int_0^{x_1} \frac{d\phi_1}{dx} E \frac{d\phi_1}{dx}\, dx$$

$$= \int_0^{x_1} E \frac{1}{h_1^2}\, dx = \frac{1}{h_1^2} \int_0^{x_1} E(x)\, dx$$

Similarly for K_{12}:

$$K_{12} = \int_0^L \frac{d\phi_1}{dx} E \frac{d\phi_2}{dx}\, dx = \int_0^{x_1} \left(-\frac{1}{h_1}\right) E \frac{1}{h_1}\, dx = -\frac{1}{h_1^2} \int_0^{x_1} E(x)\, dx$$

Pattern of Element Contributions

Computing stiffness terms like K_{13}:

$$K_{13} = \int_0^L \frac{d\phi_1}{dx} E \frac{d\phi_3}{dx}\, dx = 0$$

Since ϕ_1 and ϕ_3 have no overlapping support.

This pattern shows why the FEM matrices have a banded structure - there are only local interactions between adjacent elements.

Element-wise Assembly

We recognize that:

$$K_{ij}^g = \sum_{\text{elem}} K_{ij}^e, \quad \text{where } K_{ij}^e = \int_{\Omega_e} \frac{d\phi_i}{dx} E \frac{d\phi_j}{dx}\, dx$$

$$F_i^g = \sum_{\text{elem}} F_i^e, \quad F_i^e = \int_{\Omega_e} \phi_i f\, dx + t^* \phi_i|_{\Gamma_t \cap \Omega_e}$$

This element-wise assembly is key to FEM efficiency.

Reference Element Transformation

To make calculations systematic, we define a master (reference) element in a local coordinate system $\zeta \in [-1, 1]$.
We need a mapping from reference to physical coordinates:

$$x = \sum_{i=1}^{2} \chi_i \hat{\phi}_i(\zeta) = M_x(\zeta)$$

where:

- χ_i are the physical coordinates of the nodes
- $\hat{\phi}_i(\zeta)$ are the shape functions in reference coordinates

Shape Functions in Reference Space

For linear elements in 1D:

$$\hat{\phi}_1 = \frac{1}{2}(1 - \zeta) \quad \text{and} \quad \hat{\phi}_2 = \frac{1}{2}(1 + \zeta)$$

Differential Relations

We need the mapping between derivatives:

$$\frac{d}{d\zeta} = \frac{dx}{d\zeta}\frac{d}{dx} = J\frac{d}{dx}$$

And the inverse relation:

$$\frac{d}{dx} = \frac{d\zeta}{dx}\frac{d}{d\zeta} = \frac{1}{J}\frac{d}{d\zeta}$$

Where $J = \frac{dx}{d\zeta}$ is the Jacobian of the transformation.

Numerical Integration: Gaussian Quadrature

For numerical integration, we use Gaussian quadrature:

$$\int_{x_i}^{x_{i+1}} F(x)\,dx = \int_{-1}^{1} F(x(\zeta))J(\zeta)\,d\zeta \approx \sum_{q=1}^{G} w_q F(x(\zeta_q))J(\zeta_q)$$

Gauss quadrature can integrate a polynomial of order $2G-1$ exactly with G points.

Gaussian Quadrature Points

Gauss rule	ζ_i	w_i
2	0.5773502691896	1.0000000000000
	-0.5773502691896	1.0000000000000
3	0.0000000000000	0.8888888888889
	0.7745966692241	0.5555555555556
	-0.7745966692241	0.5555555555556

Two points are sufficient for linear elements since the integrands in K_{ij} have at most cubic terms.

Computing Element Matrices with Quadrature

The element stiffness matrix is computed as:

$$K_{ij}^{e} = \int_{\Omega_e} \frac{d\phi_i}{dx} E \frac{d\phi_j}{dx}\, dx = \sum_{q=1}^{G} w_q \left[\frac{d}{d\zeta} \hat{\phi_i}(M(\zeta)) \frac{d\zeta}{dx} \right] E \left[\frac{d}{d\zeta} \hat{\phi_j}(M(\zeta)) \frac{d\zeta}{dx} \right] J$$

Evaluated at $\zeta = \zeta_q$

Similarly for the force vector:

$$F_i^e = \int_{\Omega_e} \phi_i f\, dx = \sum_{q=1}^{G} w_q \hat{\phi_i}(M(\zeta)) f(M(\zeta)) J$$

Evaluated at $\zeta = \zeta_q$, plus any boundary terms.

Post-Processing

After obtaining the nodal values a_i, we can compute:

- **Displacements** at any point: $u(x) = \sum_{i=1}^{N} a_i \phi_i(x)$
- **Strains** at any point: $\varepsilon(x) = \frac{du}{dx} = \sum_{i=1}^{N} a_i \frac{d\phi_i}{dx}$
- **Stresses** at any point: $\sigma(x) = E(x)\frac{du}{dx} = E(x)\sum_{i=1}^{N} a_i \frac{d\phi_i}{dx}$

Implementing Boundary Conditions

Essential (Dirichlet) boundary conditions (e.g., $u(0) = u^*$):

$$\begin{bmatrix} 1 & 0 & \dots & \dots & 0 \\ K_{21} & K_{22} & \dots & \dots & K_{2N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ K_{N1} & K_{N2} & \dots & \dots & K_{NN} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ \vdots \\ a_N \end{bmatrix} = \begin{bmatrix} u^* \\ F_2 \\ \vdots \\ \vdots \\ F_N + \dots \end{bmatrix}$$

Speaker notes